Enterprise Cloud Management

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Agenda

- Introduction
- Ye Olde Cloude
- Users
- Budget
- Automation
- Summary/Crystal Ball
- Q & A
- Case Studies

Introduction

- About Me
  - Cloud engineer with Enstratius, Inc.
  - Offices in US, Scotland and New Zealand
  - Software Systems Engineer, Father, Husband, Nerd...
  - Twitter: @zomgreg

- What I do:
  - Infrastructure-as-a-Service (IaaS)
  - Enterprise cloud initiatives
  - Cloud solutions (hybrid and private cloud)

- The Format
  - The purpose of this presentation is to inform.
  - You “should”, “they are”, best practice....

Off Topic Already? Cloud management defined:

Automation	Security	Finance	Hybrid Cloud	Monitoring	Misc
Auto-recovery	File system encryption	Cost tracking	Public Clouds: AT&T Azure Cloud Central CloudSigma GoGrid Google Rejila ReliaCloud Server Express Terremark	Push notifications	On premise deployment
Non-AWS auto-scaling	Key management	Chargebacks		IDS integration	Pause servers
Cloud bursting	Custom roles	Multi-currency		MS System Center	Network definitions
Cross-cloud auto-scaling	User termination	Cross-cloud budgets		Budget alerts	Port definitions
Cross-cloud backups	Enterprise Integration: LDAP/AD SAML 2.0	Soft quotas		SSL alerts	AWS SNS
Server pause/ term. scheduling		Hard quotas	Cloud Platforms: Atmos Mezeo vCloud vSphere XenServer		Black label UI
RAID automation	IDS integration	Budget Reporting by: Budget code Resource User			CloudAudit support
Encryption automation	User VM Access: Unix/SSH Windows/RDP				
Automatic patch rollout					

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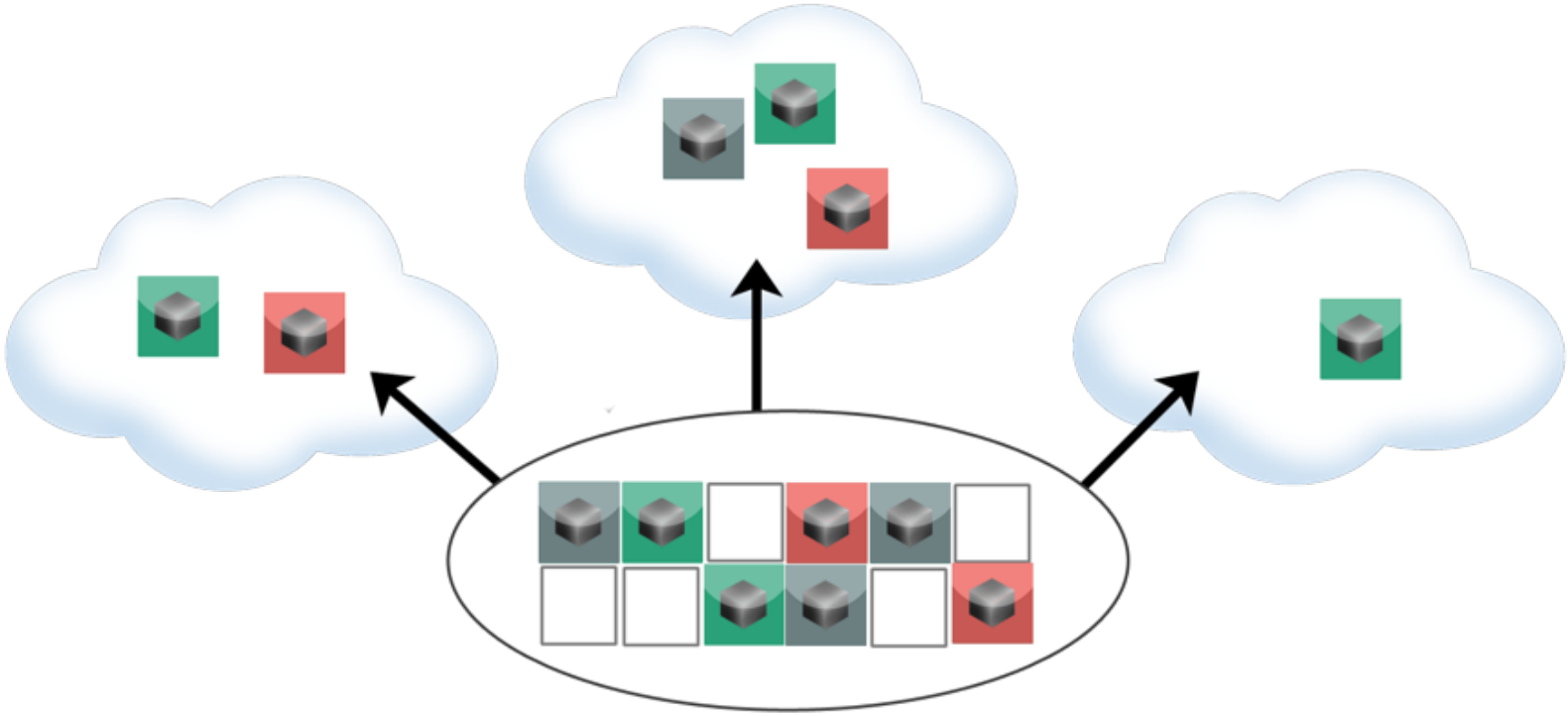
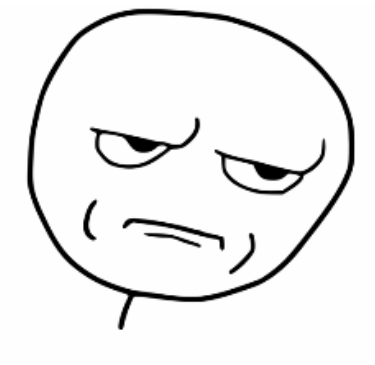
Ye Olde Cloude

- About Cloud
  - What is not cloud?
  - What is cloud?
  - Virtualization vs. Cloud



Ye Olde Cloude



"Let's be clear:

Virtualization is not cloud computing. It is a step on the journey, but it is not the destination."

Brad Anderson, Microsoft

Cloud is like honey badger. It don't give a darn about hypervisors.

George Reese, CTO, enStratus

Ye Olde Cloude

A cloud will have the following characteristics:

1. On-demand self-provisioning
2. Locational independence
3. Perception of infinite capacity

Ye Olde Cloude

Virtualization alone does not a cloud make.

Managing a virtual infrastructure requires knowledge of the physical infrastructure and new concepts like over-provisioning and clusters.

Xen Server

KVM

VMWare

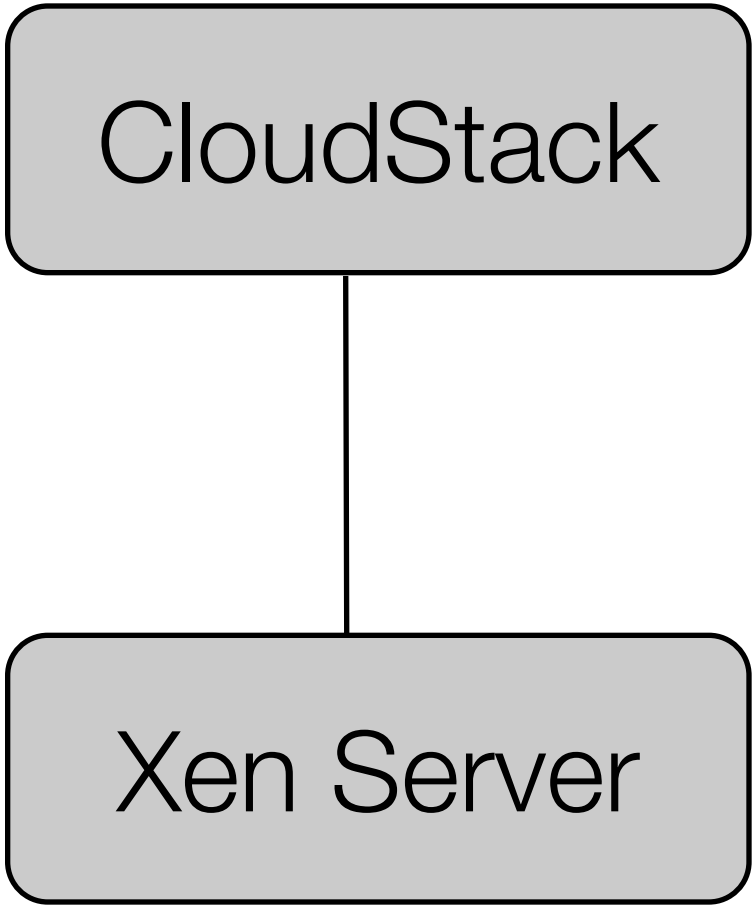
Ye Olde Cloude

Technically, this is a cloud.

BUT:

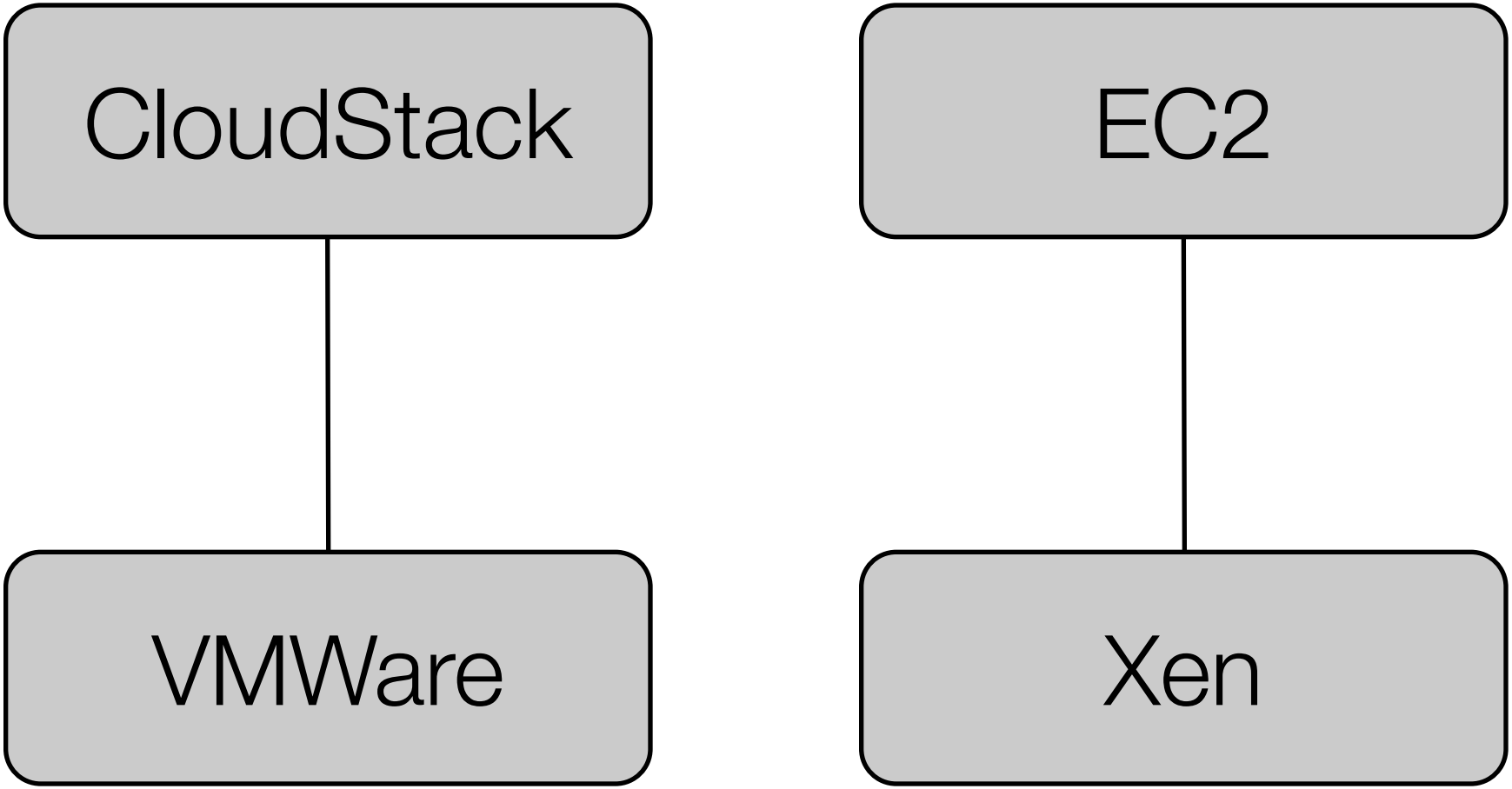
It is a single cloud w/o any external awareness beyond its own borders.

It lacks governance.



Ye Olde Cloude

This is not a hybrid cloud:



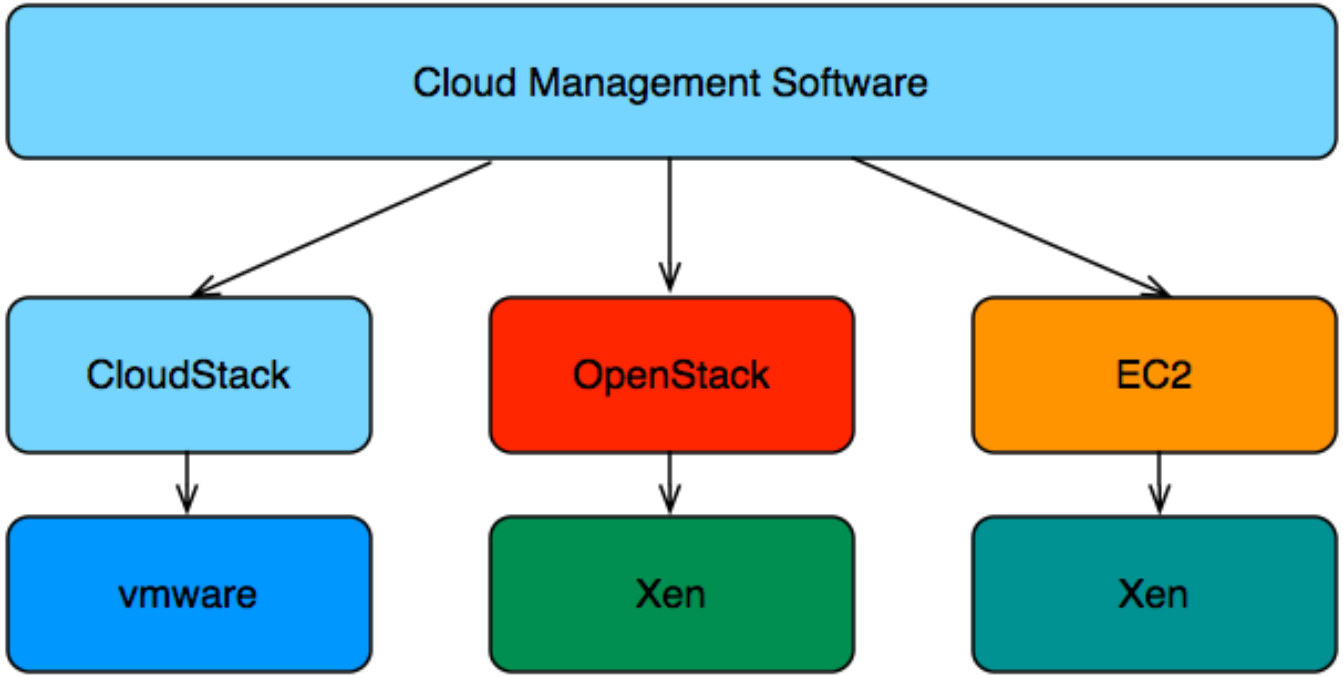
This is two clouds.

Ye Olde Cloude

Cloud management tools complete the cloud.

They operate across multiple clouds and provide the ability to manage those clouds as a unified infrastructure while supporting critical IT policies and enabling application automation.

The company that controls the cloud management layer controls the infrastructure.



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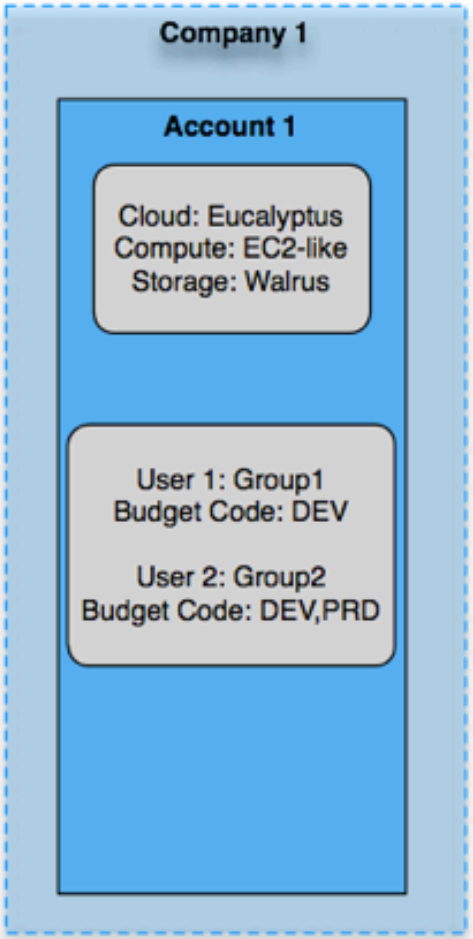
One cloud, two cloud. Red Cloud, Blue Cloud.

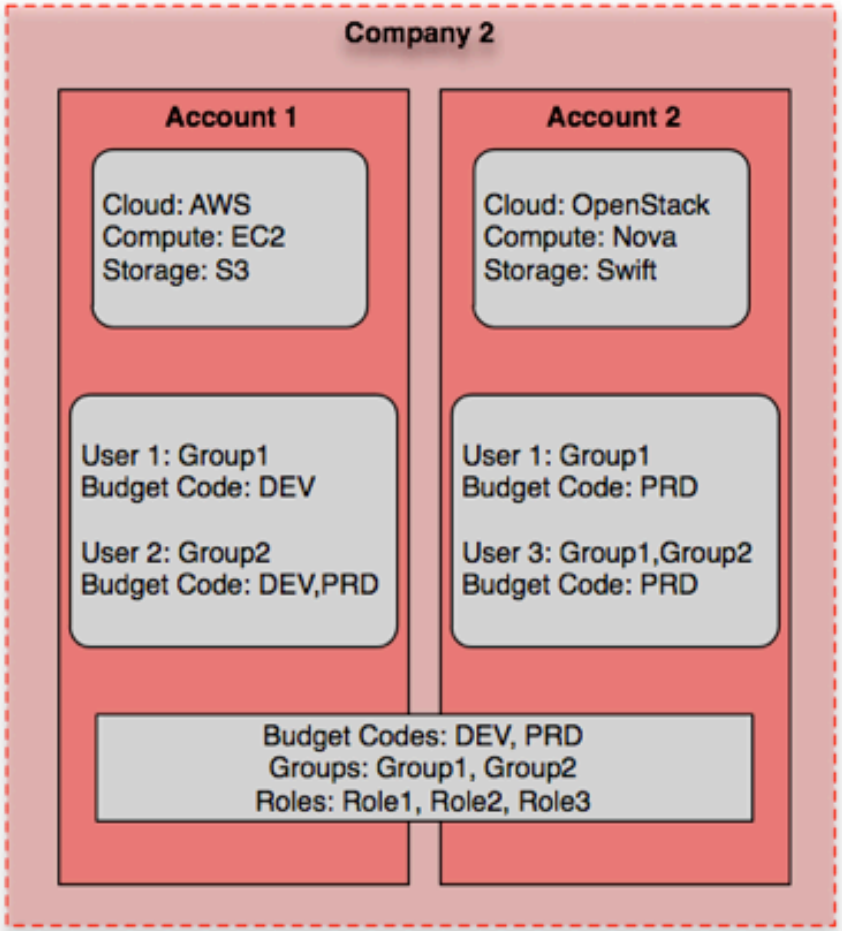
Accounts and Billing Codes

A company can have many accounts.
Accounts have only one set of cloud credentials.
Budget codes are visible across the company.
Budget codes aggregate costs across all accounts.

Users, Groups/Roles

Groups are visible company-wide, and are tied to 0 or 1 Role.
Roles are visible company-wide, and control access to resources.
Users may be assigned to multiple groups.
A group in one account can be assigned to a different role in another account.





Users. Ah, Users.

- It's a multi-cloud world.
  - Users should have customizable access to cloud resources.
- Active Directory/LDAP
  - Users should be group-able.
  - Users should be subject to Role Based Access Controls (RBAC)
- The user profile
  - SSH public key
  - RDP
  - IAM (cloud access)
- Budgets
  - User activity should be tracked with budgets.

The implementation of user controls should be seamless to the user.

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Red light, green light.

Console | Automation | Configuration Management | Compute | Network | Platform | Users | Company Settings | Account Settings | Resources

Budget Codes + Add Budget Code

Bulk actions ▸ Show 10 entries Filter:

ID	Name	Finance Code	Soft Quota	Hard Quota	Run Rate	Current Usage	Projected Usage	Budget	Active	
100	Default	DEF	$100.06	$500.29	$0.65	$273.56	$605.71		✓	actions ▸
111	R&D Budget	DEV	$200.12	$2,001.18	$0.00	$432.70	$432.70		✓	actions ▸
50234	TestBudget	CHECK	$0.01	$2.00	$0.00	$4.95	$4.95		✓	actions ▸
50000	For Testing	TEST2	$100.68	$201.37	$0.00	$34.01	$34.01		✓	actions ▸
161	Eucalyptus	EUCA	$0.00	$100.16	$0.00	$33.29	$33.29		✓	actions ▸

Budgets:

1. Customizable
2. Aggregated (cross cloud/account/organization)
3. Currency aware
4. Charge-backable (export)

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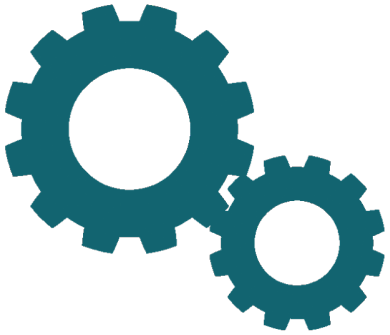
Automation

- What does automation mean?
  - Repeatable process
  - Orchestration
    - The proper ordering of a sequence of events.
    - Server launch? Multi-cloud bursting launch?
    - Software installation?
      - Kickstart? Meh.
      - Chef
      - Puppet
  - Cloud Aware
  - Application Aware

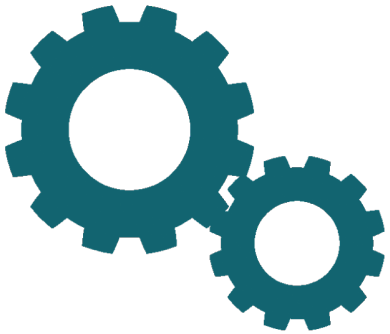
Automation Layers

1. Cloud
2. Server
3. Application
4. Governance



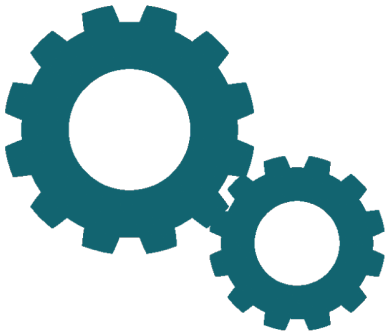
Automation: Cloud Level

- Cloud Automation addresses the following:
  - Into which cloud will a resource be started?
  - How will scaling events be triggered?
    - cross-cloud bursting?
    - hybrid cloud automation?
  - What are the limits placed on this automated environment?
    - Minimum/maximum servers
    - “Sensitivity” of scaling
    - Customized scaling
      - Application aware

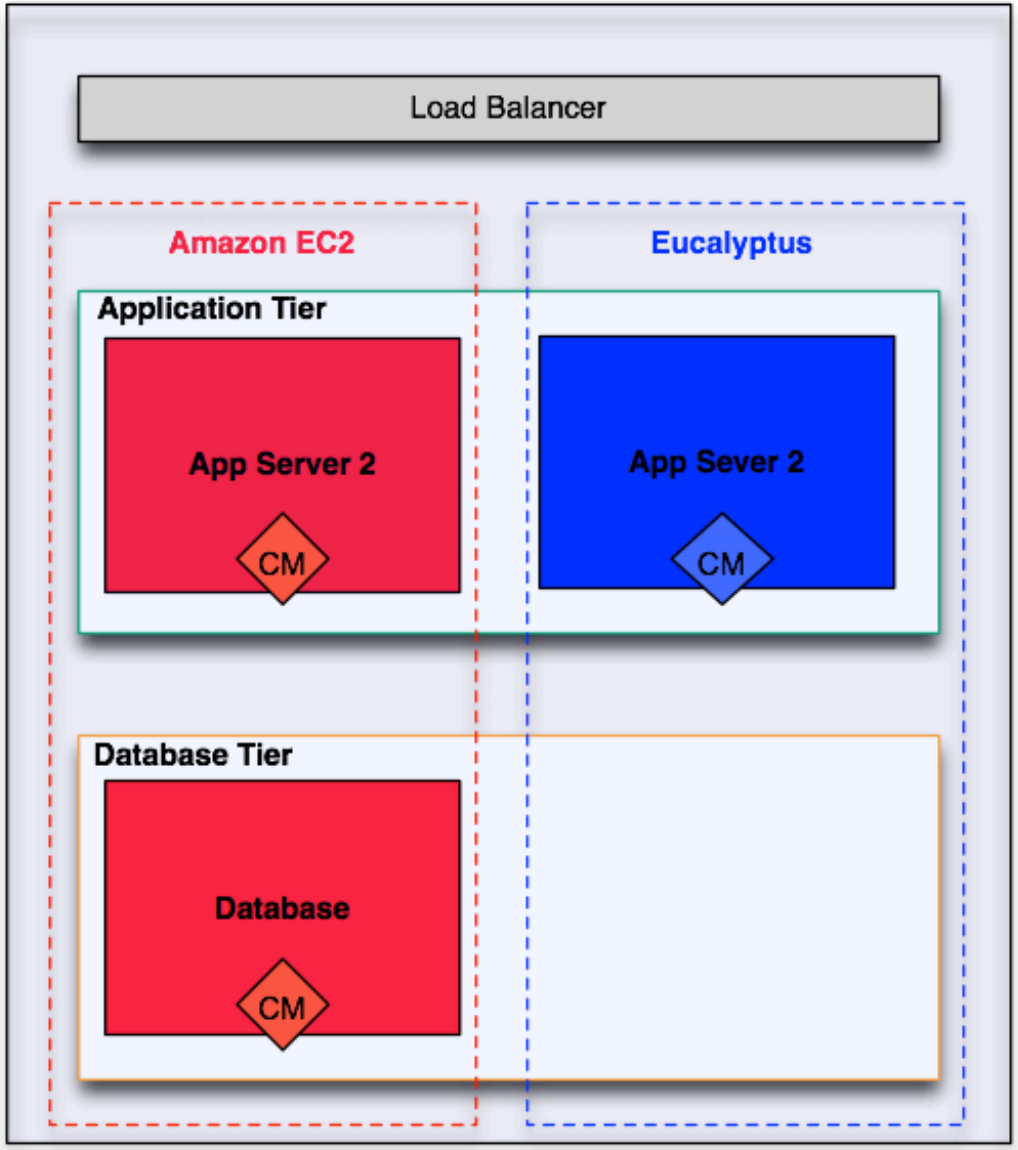


Automation: Cloud Level

- Cloud Automation: Addresses the following
  - What resources are provisioned?
    - Product size (m1.large)
    - Volumes (encrypted?)
    - IP addresses
    - Load Balancer
    - Cloud files
    - Backups (snapshots and application)
    - Firewall
    - Region, zone, data center awareness
    - Virtual Private Cloud (networking)
  - What are the relationships between cloud resources?



Automation: Cloud Level



Automation: Server Level

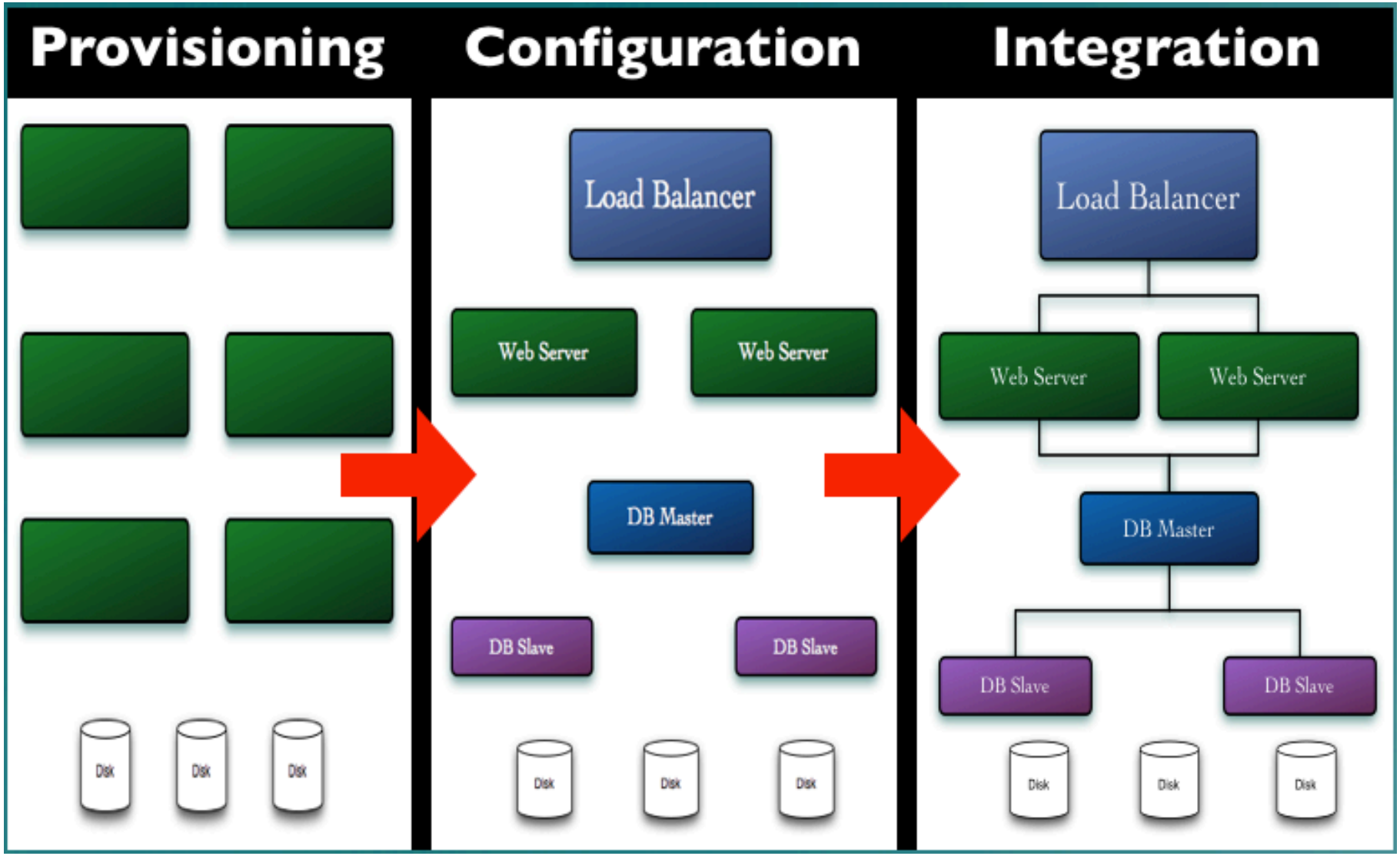
- Server automation addresses the following:
  - Just enough operating system
    - No more golden masters...
    - Seriously, no more.
  - Idempotent server builds
    - myserver.iso + δ
    - δ = configuration management client
  - Advantages?
    - Visibility
    - Audit-ability
    - Repeatability
    - Scalability
      - Thousands of nodes? One change to rule them all.

My preciousssss...

Configuration management turns servers into code.

http://www.sxc.hu/photo/452432, http://www.sxc.hu/help/7_2

Automation: Server Level



Automation: Application Level

- Application automation addresses the following:
  - Handling emphemerality
    - Servers not working? Waste it. Start a new one.
    - Cloud: The ultimate chaos monkey?
  - Application Flexibility
    - Unpredictable IP addressing
    - Automated password generation
    - Automated firewall modification
  - How to?
    - Know thy application
    - Separate credentials
    - Leverage elasticity

Code/architect as if the server upon which your application runs could leave this world at any time and be replaced.

Automation: Governance

- Automation governance means:
  - Jurisdiction awareness
    - EU/U.S.
    - HIPAA
    - PCI
  - Cloud capability awareness
    - SLA (how many 9's please?)
      - Asserted vs. Emprical (historical)
      - Real time adjustments
  - Application awareness
    - Best cloud for heavy I/O driven apps
    - Best cloud for files distribution
  - Catalog integration
    - Create your own
    - 3rd party integration

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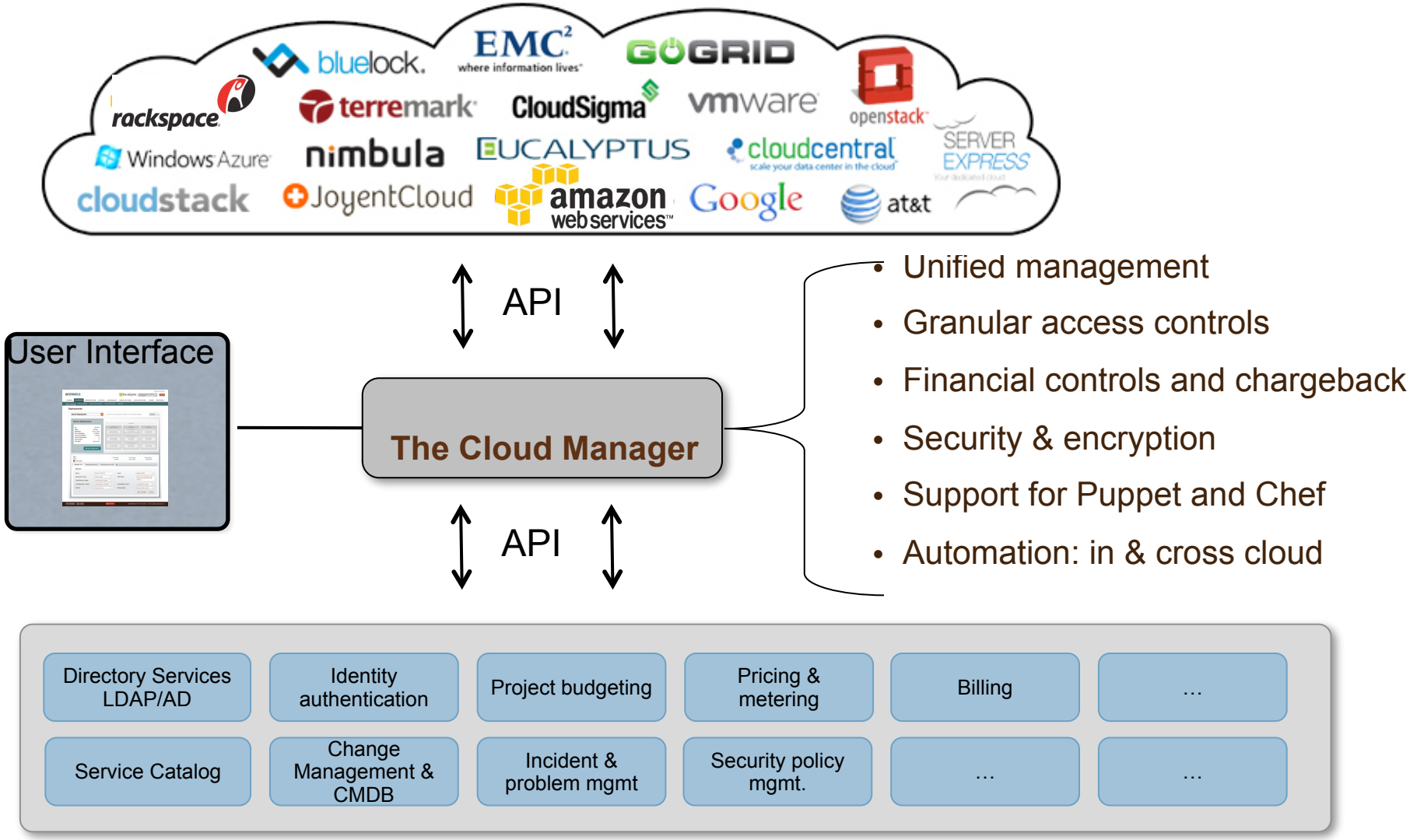
Summary


bluelock.
EMC²
where information lives
GOGRID
rackspace
terremark
CloudSigma
vmware
openstack
Windows Azure
nimbula
EUCALYPTUS
cloudcentral
SERVER EXPRESS
cloudstack
JoyentCloud
amazon web services
Google
at&t
API
User Interface
The Cloud Manager
API
• Unified management
• Granular access controls
• Financial controls and chargeback
• Security & encryption
• Support for Puppet and Chef
• Automation: in & cross cloud
Directory Services LDAP/AD
Identity authentication
Project budgeting
Pricing & metering
Billing
…
Service Catalog
Change Management & CMDB
Incident & problem mgmt
Security policy mgmt.
…
…

Summary

1. Pay your technical debt up front.
2. Go multi-cloud and integrate.
3. Employ a cloud-governance tool.
4. Employ configuration management tools for idempotent server process operations.
5. Build a workflow.
6. Track your costs.

Summary

7. Use the cloud
8. Engage for POC
9. Choose an application
10. Leverage cloud management.

The Crystal Ball

Disclaimer: Pure speculation ahead!

- Maturation of private cloud technologies.
  - Closing the feature gap on EC2
  - Stability of installation, ease-of-deployment
- Multi-cloud becomes commonplace
- Emphasis on cloud integration/independence.
- DIY Platform-as-a-Service
  - Cloud Foundry
- SaaS managing IaaS managing PaaS.
  - Meta-cloud management is already here.
  - Try not to divide-by-zero.

- Nice to haves:
  - Cold fusion, free energy, world peace.